

Chandrayaan-3 Mission Report

Indian Space Research Organisation (ISRO) — 2023

1. Mission Overview

Chandrayaan-3 is India's third lunar exploration mission and the successor to Chandrayaan-2. The mission objective is to demonstrate safe and soft landing on the lunar surface, rover roving on the Moon, and in-situ scientific experiments. The mission was launched on 14 July 2023 aboard the LVM3-M4 launch vehicle from Satish Dhawan Space Centre (SDSC) SHAR, Sriharikota. The spacecraft successfully achieved a soft landing near the lunar south pole on 23 August 2023, making India the fourth country to achieve a soft lunar landing and the first to land near the south pole.

The total mission cost was approximately INR 615 crore (approximately USD 75 million), making Chandrayaan-3 one of the most cost-effective lunar missions in history. The mission demonstrated ISRO's capability for interplanetary exploration and advanced autonomous navigation.

2. Mission Architecture

2.1 Propulsion Module

The Chandrayaan-3 propulsion module carries the spacecraft from launch injection to lunar orbit. It uses a 440 N liquid apogee motor (LAM) as the main engine for orbit-raising manoeuvres and lunar orbit insertion. The propulsion module has a total mass of 2148 kg including 1696.39 kg of propellant. Four 50 N thrusters provide attitude control. The propulsion module also carries the SHAPE (Spectro-polarimetry of HAbitable Planet Earth) scientific payload, which studies Earth's spectral and polarimetric signatures from lunar orbit.

2.2 Lander Module (Vikram)

The lander module, named Vikram, has a mass of 1752 kg including the Pragyan rover. It is equipped with four throttleable engines of 800 N thrust each for the powered descent phase. The lander uses a Lander Hazard Detection and Avoidance Camera (LHDAC) and a Laser Doppler Velocimeter (LDV) for autonomous hazard detection during the final descent. Solar panels generate 738 W of power. The lander communicates with IDSN (Indian Deep Space Network) at Byalalu via S-band and X-band links, and also uses the Chandrayaan-2 orbiter as a communication relay.

2.3 Rover (Pragyan)

The Pragyan rover has a mass of 26 kg and is designed for one lunar day (approximately 14 Earth days) of operation. It moves at a speed of 1 cm/s on the lunar surface. The rover carries two scientific payloads: APXS (Alpha Particle X-ray Spectrometer) for elemental analysis of the lunar surface, and LIBS (Laser Induced Breakdown Spectroscope) for determining the elemental composition of rocks and soil. The rover generates 50 W of power from its solar panel.

3. Landing Site

The Chandrayaan-3 lander targeted a landing site near the lunar south pole at approximately 69.37 degrees South latitude and 32.35 degrees East longitude. The south polar region is scientifically significant because permanently shadowed craters in this region may contain water ice deposits. These water ice deposits could serve as resources for future lunar exploration and potential human habitation on the Moon. The landing site was selected based on orbital imagery from Chandrayaan-2 and other lunar orbiters.

4. Scientific Discoveries

The Pragyan rover confirmed the presence of sulphur on the lunar south pole surface for the first time through in-situ measurements using the LIBS instrument. The APXS instrument detected aluminium, calcium, iron, chromium, titanium, manganese, silicon, and oxygen in the lunar soil. The ChaSTE (Chandra's Surface Thermophysical Experiment) instrument on the lander measured the temperature profile of the lunar surface and found a significant temperature difference of approximately 60 degrees Celsius between the surface and 8 cm below the surface.

5. Communication System

Chandrayaan-3 uses a dual-frequency communication system. The lander communicates directly with the Indian Deep Space Network (IDSN) ground station at Byalalu, Bengaluru using S-band (uplink) and X-band (downlink) frequencies. Additionally, the Chandrayaan-2 orbiter, which remains operational in lunar orbit, serves as a communication relay between the lander and Earth. This relay capability was pre-planned as a contingency and was successfully demonstrated during the mission.